Roll. No.: _____

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science and Engineering
Mid-Semester Test, March 2026

Course Name: **Quantum Computing** | Course Code: **UCS619**
Total Marks: **30** | Faculty: **NEG, SAM, VAP**
Total Time: **120 Minutes** | Date: **9/3/26**

Note: Attempt all questions. Subparts of each question must be solved in one place.

1. **(a)** A quantum system is in the state **(3)**

$$|\psi\rangle = \frac{3i\,|0\rangle + 4\,|1\rangle}{4}$$

(i) Is this a valid qubit state? Calculate.
(ii) Express the state in the $\{|i\rangle, |-i\rangle\}$ basis.
(iii) Express the state in the $\{|+\rangle, |-\rangle\}$ basis.

(b) Find $X \otimes Y\,|\psi\rangle$, where **(2)**

$$|\psi\rangle = \frac{|0\rangle\,|1\rangle - |1\rangle\,|0\rangle}{\sqrt{2}}$$

2. **(a)** An operator acts on the qutrit basis in the following way: **(3)**

$$A\,|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad A\,|1\rangle = |0\rangle, \quad A\,|2\rangle = |1\rangle$$

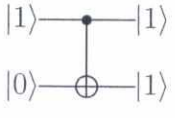
Find the expected value of the operator A for the state

$$|\psi\rangle = \frac{1}{2}\,|0\rangle - \frac{i}{2}\,|1\rangle + \frac{1}{\sqrt{2}}\,|2\rangle$$

(b) Find the trace of the following operator A expressed in the $|0\rangle, |1\rangle$ basis: **(2)**

$$A = (4i+1)\,|0\rangle\langle 0| + 3\,|0\rangle\langle 1| - 2\,|1\rangle\langle 0| + (2i+1)\,|1\rangle\langle 1|.$$

3. **(a)** Consider the CNOT gate with inputs as shown in the following diagram, where it creates a copy of the $|1\rangle$ state. Does this violate the no-cloning theorem? Explain why or why not. A precise 2-3 line answer is sufficient. **(1)**



(b) What do you mean by superdense coding? Explain its working with a complete quantum circuit. **(4)**

4. **(a)** Show that the CCNOT gate is its own inverse. **(2)**

(b) Define $\Pi_a = \frac{1}{2}(I - X)$ and $\Pi_b = \frac{1}{2}(I - Z)$. Show that Π_a and Π_b are projection operators, i.e., $\Pi^2 = \Pi$. Here, X and Z are Pauli operators. **(2)**

(c) Given $|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, Calculate $\langle\psi|\Pi_a \otimes \Pi_b|\psi\rangle$. **(1)**

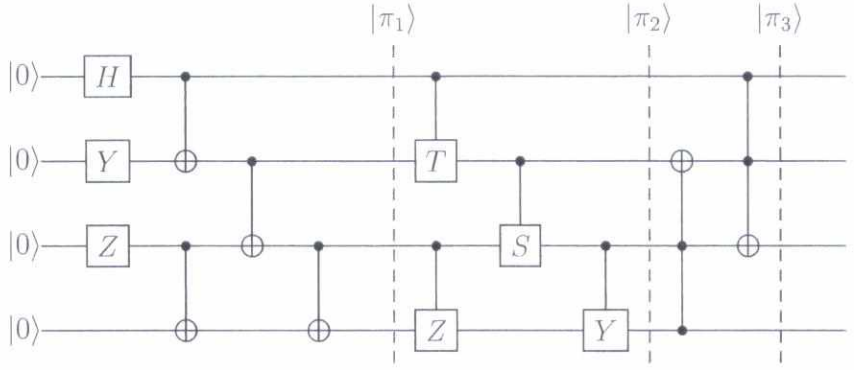
5. Consider $M = a\,|+\rangle\langle+| + b\,|-\rangle\langle-|$ and $\psi = \cos\alpha\,|00\rangle + \sin\alpha\,|11\rangle$ where $a, b, \alpha \in \mathbb{R}$. **(2+2+1)**

   (a) Show that M is an observable.

   (b) When the first qubit is measured with M, what is the probability that the measurement finds the first qubit in the state $|+\rangle$? What is the state of the system after measurement?

   (c) Compute the expected value of M: $\langle\psi|M|\psi\rangle$.

6. Consider the following quantum circuit. Calculate the intermediate states $|\pi_1\rangle$, $|\pi_2\rangle$, and the final state $|\pi_3\rangle$. **(2+2+1)**



End of the Question Paper

Question-wise COs and Bloom's taxonomy

Q.1	Q.2	Q.3	Q.4	Q.5	Q.6
CO1, L2	CO1, L2	CO4, L3	CO2, L4	CO1, L3	CO2, L3, L5